

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination- 2024

Course: B. Tech.

Semester : VII

Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/

Electrical & Electronics Engg. / Electrical & Power Engineering

Subject Code & Name: BTEEPE703A Energy Audit and Conservation

Max Marks: 60

Date: 05/07/2024

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|--|---------------|---|
| A) Explain UNFCCC in details. | Remembering | 6 |
| B) Explain importance of energy conservation in India. | Understanding | 6 |
| C) State classification of energy sources with one example for each. | Understanding | 6 |

Q.2 Solve Any Two of the following.

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|--|---------------|---|
| A) Define the energy audit as per the energy conservation Act 2001 and Need for energy audit? | Understanding | 6 |
| B) Explain following instruments used in Energy Audit with their application: i) Combustion Gas Analyzer ii) Lux Meter | Understanding | 6 |
| C) Explain Various Energy conservation opportunities in Air conditioning system. | Remembering | 6 |

Q. 3 Solve Any Two of the following.

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|---|---------------|---|
| A) Enlist energy conservation techniques in Heating system. | Understanding | 6 |
| B) Why it is important to reduce Transmission and Distribution losses? Discuss various methods for reducing technical losses. | Understanding | 6 |
| C) Explain the energy conservation in motive power applications. | Understanding | 6 |

Q.4 Solve Any Two of the following.

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|---|---------------|---|
| A) Describe demand side management in Domestic Sector. | Understanding | 6 |
| B) Write a note on Demand side management. | Understanding | 6 |
| C) Explain supply side concept with advantages and disadvantages. | Remembering | 6 |

Q. 5 Solve Any Two of the following.

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|---|----------------------|----------|
| A) Write a note on: i) Simple payback period ii) Return on Investment (ROI) | Understanding | 6 |
| B) Explain Implementation and Monitoring of Energy Conservation Projects. | Understanding | 6 |
| C) What is NPV of a project, (life 3 years) which requires an investment of Rs. 50000 and yield Rs. 30000 in the 1st year and Rs. 40000 /- in the next year, if the interest rate is 10%. | Understanding | 6 |

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